

Bibliometric Analysis of Global Research Trends in Predicting Financial Distress among Small and Medium-Sized Enterprises (SMEs): A Systematic Review

 Aliyu Abubakar;  Ibrahim Mahmoud; &

 Abdulsalam Kabir Tunga

Department of Accountancy, Waziri Umaru Federal Polytechnic, Birnin Kebbi.

Corresponding Author: abubakaraliyu132@gmail.com

DOI: <https://doi.org/10.70382/hujhrms.v10i7.040>

Keywords: Financial distress prediction, Small and Medium-Sized Enterprises (SMEs), Bibliometric analysis, Global research trends, Policy implications

Abstract

The rise in financial distress has become a critical concern for researchers worldwide, prompting efforts to develop predictive models that mitigate bankruptcy risks among businesses. Such financial instability threatens a company's reputation and performance, often leading to volatility, uncertainty, and potential insolvency. This study conducts a bibliometric analysis to examine global research trends in financial distress prediction among Small and Medium-Sized Enterprises (SMEs). Using a systematic review of peer-reviewed articles published from 2014 to 2024, sourced through the Google Scholar database, this analysis evaluates publication patterns, research productivity, and key themes in the literature. The findings reveal a limited volume of global studies on SME financial distress, with particularly sparse research in Nigeria, underscoring a critical gap in regional understanding. Results indicate an urgent need for policy interventions that prioritize research funding, support for predictive analytics, and strategic frameworks to strengthen SME resilience. The study recommends further bibliometric analysis across multiple

databases to deepen insights into financial distress prediction and support evidence-based policy development for SMEs.

Introduction

The emergence of financial distress has forced a significant challenge to the researchers globally to extensively develop models for predicting company bankruptcy in order to address the financial crises face by the business organizations that are capable of jeopardizing their image and performance which is characterized by pronounced financial instability with the potential risk of volatility, uncertainty, and bankruptcy. Financial distress refers to the period of accumulated difficulties, erupted tensions and conflict. Financial distress refers to as costly event that has the possibility of determining the firms' optimal capital structures. Financial distress is considered costly because it creates a tendency for organizations to act in a way that are harmful to nonfinancial stakeholders and debtholders and weakened condition that induces an aggressive response by competitors that enhance an opportunity to gain market share. Financial distress defines in terms of insolvency, or the inability to pay obligations as at when due (Scott, 1981; and Bahnson and Bartley, 1992).

Financial distress can force managers to make difficult value-maximizing choices, thereby improving firm values which they would otherwise avoid on the other hand financial distress can cause significant losses in some cases and motivate value-maximizing choices in others. The overall cost and benefit of financial distress is quite difficult to quantify as it benefits firms by forcing efficient operating changes,

Researchers argued on the techniques for measuring financial distress in which the use of legal bankruptcy as measure of financial distress was considered by scholar for example (see Dietrich, 1984; Delaney, 1991; and Bahnson and Bartley, 1992) and was criticized underground that bankruptcy is a legal definition rather than economic event by Delaney (1991), who argued that a company can be economically solvent even though legal recognition of bankruptcy occur. Consequently, using a legal event such as bankruptcy for developing statistical models in testing the predictive financial distress may produce results not consistent with economic reality leading to improper conclusions.

In order to provide a comprehensive understanding on the literature on Financial Distress among Small and Medium-Sized Enterprises (SMEs) this study aims at providing comprehensive bibliometric analysis of global research trends on financial distress prediction among Small and Medium-Sized Enterprises (SMEs). The analysis will cover peer-reviewed articles published in English-language academic sources over the past decade (2014-2024). The study will employ bibliometric tools and techniques, such as citation analysis, co-authorship analysis, and keyword analysis, to identify emerging trends, patterns, and gaps in the existing literature.

Research Questions:

RQ1. What is the publication productivity of the research on Financial Distress Prediction of Small & Medium Enterprises?

RQ2. What are the most influential articles on Financial Distress Prediction of Small & Medium Enterprises?

RQ3. What are the most prominent topic and themes on Financial Distress Prediction of Small & Medium Enterprises?

The Concept of Financial Distress

In 1966 a study conducted on business failure, corporate financial risk and bankruptcy served as the foundation for the studies of financial distress analysis, who does not exactly define financial distress but in his study sample 79 companies in which 59 were on the basis of bankruptcy, 3 were unable to pay their debts as the date of maturity, while 16 were required to pay dividends on preferred shares (Wu Shinong & Lu Xianyi, 2001). Inherently Beaver defines financial distress to include inability to pay dividends, non-payment of interest and principal on debt and or bonds as serious financial distress.

Current ratio was added in 1968, by Altman as an indicator of financial distress considering it as a symptom of poor cash flow. Similarly, Ohlson (1980) project that bankruptcy timing had a significant impact on the prediction of financial distress. Zmijewski (1984) argued that taking large sampling of financially distressed businesses have a severe effect on the accuracy of financial distress predictions.

Financial distress or failure model was developed by Altman (1968) using seven ratios i.e. retained earnings to total assets, working capital to total assets, market value equity to the book value of total debt, earnings before interest, taxes to total assets, and sales to total assets in predicting the financial failure. On the other hand, Zmijewski, M.E. (1984) used nine financial variables to develop a financial distress prediction model using return on assets (ROA) and liquidity as financial distress prediction indicators, employing the sample of 400 operating enterprises in of good standing with 40 insolvent companies as research subjects.

Several studies attempted to different financial distress of an entity with no specific or unanimous or general agreement as to financial distress indicator which inform the need for the systematic and review to address the identify literature gap by providing bibliometric and systematic review of intellectual literature on financial distress prediction to provide an in depth understanding of scholarly article on financial distress by providing answers to the question as to the publication productivity, most influential articles and the most prominent topic and themes on Financial Distress Prediction of Small & Medium Enterprises for the period of 10 year from 2014 to 2024 which to the best knowledge of the researcher was not conducted before.

Bibliometric analysis

Bibliometrics analysis is the application of mathematical and statistical method first produced by Pritchard (1969). It applied quantitative analysis of existing empirical published literature for assessing the patterns of publication within a particular area of field of study (De Bellis, 2009). Bibliometrics analysis enables the researchers in examining the body of literature in their area of interest through identify major themes (Grant et al., 2000; Vogel and Güttel, 2013). Bibliometric analysis combined science mapping techniques to visualize the intellectual structure of a researcher area of study interest (Cobo et al., 2011). The Analysis consists of several

methods depending on the different information used in the Bibliometrics data set (Van Raan, 2005).

The most bibliometric analysis includes citation-based analysis, keyword co-occurrence analysis or co-word analysis, and co-authorship collaboration analysis (VanEck and Waltman, 2014). In the citation-based analysis, further classifications are bibliographic coupling, citation analysis, and co-citation analysis, (Van Eck and Waltman, 2014).

Co-citation analysis is the most widely used bibliometric analysis method (Ding et al., 2001), and is define to mean two publications which are cited together in one article (Small, 1973). Suggesting that, when two article published are frequently co-cited by the other authors, it is an indication that these two references have something in common (Benckendorff and Zehrer, 2013). Co-citation analysis is used to discover the clusters of co-citation pairs, thereby enhancing scholars to obtain an insights knowledge of the cumulative tradition, and intellectual structure of scientific research articles (Small, 1978; Culnan, 1986; Pasadeos et al., 1998). Co-citation analysis has been considered as the superior in showcasing disciplinary structures compared to other bibliometric analysis methods (Bichteler and Eaton, 1980; Chang et al., 2015), yet it has not been able to provide a content picture of the research topics dealt with in the literature. On the other hand, co-was developed to address this kind of analytical problem (Callon et al., 1991). Co-word analysis is done on the bases of the frequency of co-occurrence of author's keywords used in the reviewed studies that is, the number of papers in which two keywords appear together (Whittaker, 1989). This will measure the strengths of the keyword co-occurrence links, co-word analysis visualizes and reveals the association between keywords (Callon et al., 1991; Su and Lee, 2010). Keywords are the terms used to demonstrate the core of a research article, co-word analysis is often used to explore the concept network of research topics and trends in a specific discipline (Callon et al., 1991; Ding et al., 2001). However, term changes over time posted in stability weakness to co-word analysis (Leydesdorff, 1997; Astrom, 2002).

In order to address the weakness associated with various bibliometric analysis methods this review aims at combining various methods to explore research trends in financial distress predictions while previous studies using a combination of cocitation and co-word analyses this study will employe systematic review and biblometrics analysis to indicated complimentary roles of the two method analyses as suggested by Astrom, 2002 that, the combination of the two methods will provided better results. Similarly, Chang et al. (2015) suggested combining co-citation analysis with keyword analysis could not only create a comprehensive map of the intellectual structure of certain discipling, but enhances a deeper understanding of its development.

Research methodology

The study collected data base on the study purpose and the research questions, the research methodology can be structured into 4 steps(choosing the databases for analysis; setting the time frame for the research; choosing the software for analysis and data analysis) using prepared reporting items for systematic review and meta-analysis (PRISMA see fig. 3.1) with **Search strings:("financial distress prediction" OR "financial failure prediction") AND ("agricultural SMEs" OR "small and medium-sized enterprises")**using google scholar data

bases. The choice of the data base is informed by its accessibility and complexity in terms of publication that cut across all the data bases.

The data obtained will then be extracted in two formats(Refman and CSV) for analysis Authorship co-citation using VOS view and Pivot Table &Chart in excels, on the topic **Financial Distress Prediction of Small & Medium Enterprises**. This methodology designed is consistent with (Peralungal & Natchimuthu, 2024)The bibliometric analysis of the literature allows the management of all existing studies, thus resulting in a very complex and extensive research (Melega, 2022; Grosu et al. 2022).

Identification of studies via google scholar data base for Biometric Analysis

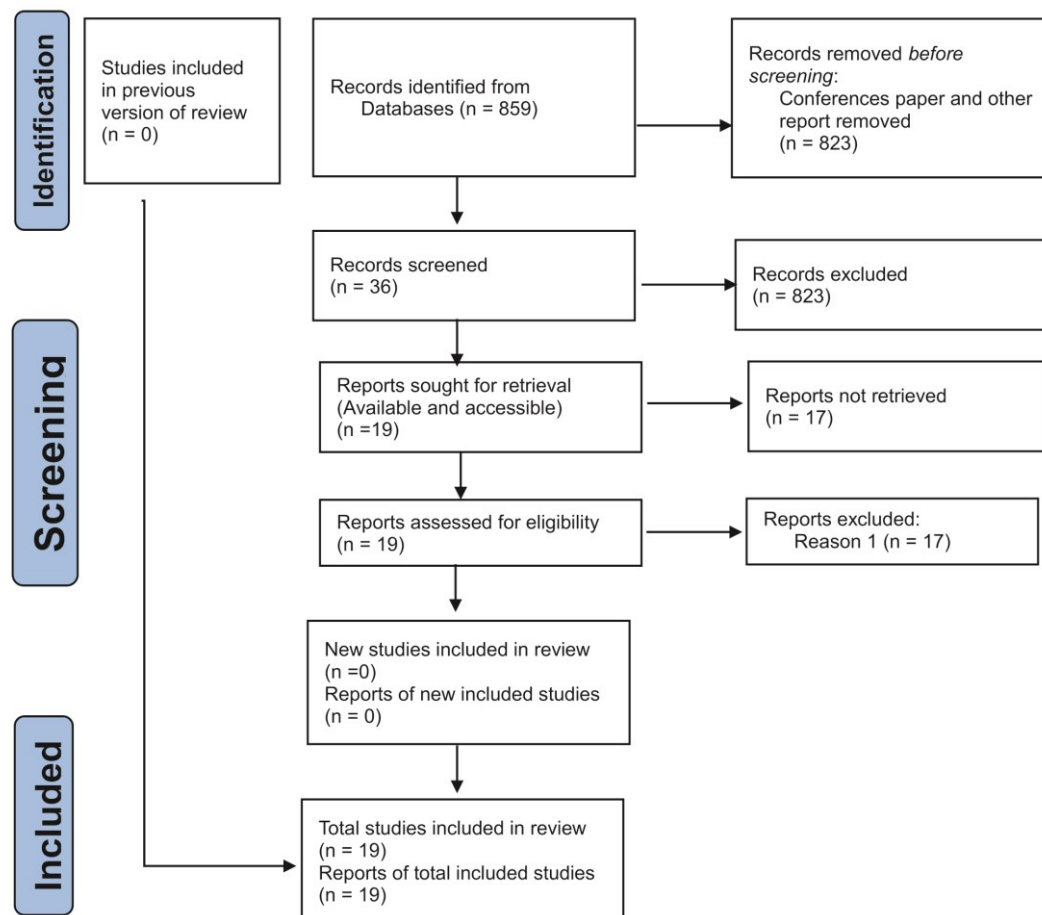


Table 3.1 PRISMA Statement and workflow for Bibliometric Analysis (2024)

Citation Analysis

The overall article returns from the google scholar database on the general search strings ("financial distress prediction" OR "financial failure prediction") AND ("agricultural SMEs" OR "small and medium-sized enterprises") on Sunday 13th October 2024 returns a total number of 859 financial distress prediction related article before screening and removing 823 document comprises of report, and conferences among others, leaving only reviewed article

writing in English language totaling to 38 article which are further screened base on accessibility and availability of the full article leaving on 19 article with for the review.

The three most cited journal in the area of financial distress studies are springer with total citations count 152 in the 2020, followed by Mdpi with citations count of 93 and Elsevier with total citation count of 90 in 2019 see (Fig 4.3)

Table 4.1 Distribution of most prominent Authors base on citations/year and title

Authors	Year of Publications	Number of Cites	Title
A Gupta, V Dengre, HA Kheruwala, M Shah	2020	151	Bankruptcy prediction for private firms in developing economies: a scoping review and guidance for future research
B Prusak	2018	93	Corporate default predictions using machine learning: Literature review
J Doering, R Kizys, AA Juan, A Fito, O Polat	2019	90	SME default prediction: A systematic methodology-focused review
H Kim, H Cho, D Ryu	2020	76	Review of research into enterprise bankruptcy prediction in selected central and eastern European countries
D Vezanzones, E Severin	2021	66	Corporate failure prediction models in the twenty-first century: a review
AI Marqués, V García, JS Sánchez	2020	55	Comprehensive review of text-mining applications in finance
F Zambrano Farias, MC Valls Martinez...	2021	31	Explanatory factors of business failure: literature review and global trends
FR Matenda, M Sibanda, E Chikodza...	2021	26	Metaheuristics for rich portfolio optimization and risk management: Current state and future trends
J Zhao, J Ouenniche, J De Smedt	2024	12	Survey, classification and critical analysis of the literature on corporate bankruptcy and financial distress prediction
H Cheraghali, P Molnár	2023	9	Ranking-based MCDM models in financial management applications: analysis and emerging challenges

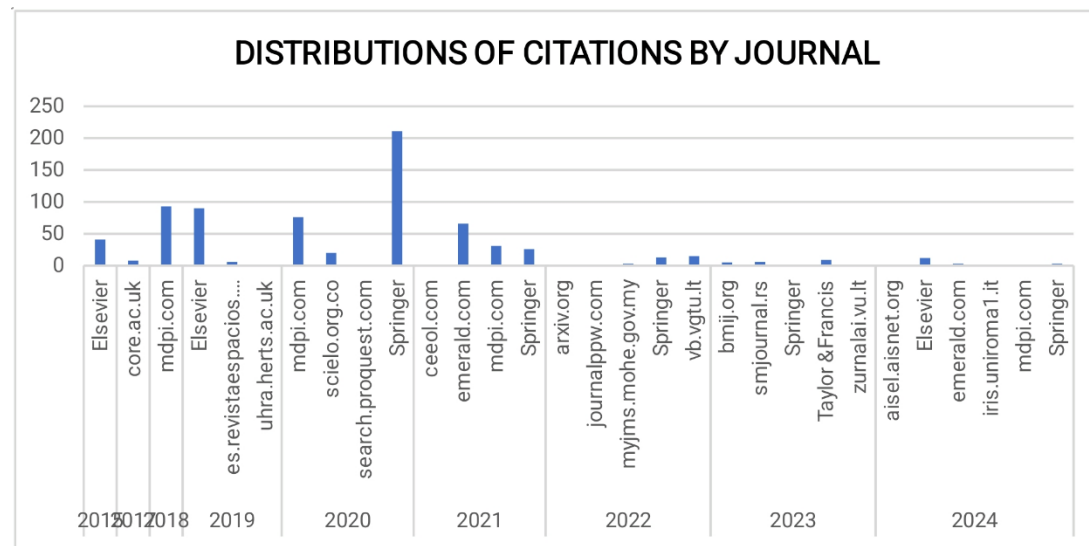


Fig. 4.1 Distributions of journal base Citations

Co-citation network

The number of articles selected as a sample in this study base on PRISMA guideline selection process, are 19 with total number of using VOSview to determine the co-citation of authors from the available 19 document. The study consider authors cited at least minimum of 3 and above citations per documents only 3 author's meet the threshold which include: Andres, Jde, Tamra, Al and Villegas, GC all with 2 document and total link strength of 4 respectively see fig. 4.2

Fig 4.2 Screen shot of Co-authorship analysis of Financial Distress Studies (VOS viewer output)

Keyword analysis and co-word network

This study examines the title from the documents gathered based on the number of occurrences and co-occurrences using VOSviewer. Specifically, this analysis used a full counting method in the constructions of the co-occurrence network. In order to explore the changes of research themes in financial distress predictions from 2014-2024 the study collected all the keyword from the title and abstract of the extracted document base on the study sample of 19 articles for conducting keyword and co-word networks analysis using VOSviewer to show the relationships among the keywords in the study area. In order to achieve that the visualized co-network was created to show the correlation between the keyword on financial distress predictions.

The size of the nodes indicates the weight of the occurrence of the terms while the thickness of joining lines indicates the strength of the relationship amongst the terms. Related words, as indicated by the same color are frequently co-occurred together (see fig 4.3) below:

Fig. 4.3 Screen shot of Keyword co-occurrence of Financial Distress Studies base on Title (VOS viewer output)

The author keywords network above mapped and presents the authors keywords as suggested by (Sweileh et al., 2017) in which color, size of circle, size of font, and thickness of connecting lines were used to show the extent of relationship of keyword to other keywords. Based on the analysis, five clusters were identified. The first cluster, which is colored red has total link strength 118 and in 26 links with total occurrence 148 containing items related to Model, Application, Artificial Intelligence, Bankruptcy, Bankruptcy prediction, Business failure, financial distress, and financial distress predictions. The second cluster, colored in green has the total link strength of 108 in 29 links with total occurrence 89 containing items related to Adaption, Challenges, Economic growth, financial performance, sustainable development focusing in Africa, and Ghana. The third cluster colored blue has total links strength 70 in 19 links with 76 total occurrences focusing on items such as Economic empowerment, poverty alleviation, Islamic microfinance in Nigeria. The fourth cluster colored yellow has the total links strength of 129 in 19 links with total occurrence of 84 with the firm performance, Poverty, in focusing in Ethiopia and Sub-saharan regions.

This provides a direction as to choice of area of study by the future researchers in consideration of the cluster and keyword with small links and less occurrence which cluster 3 and 4 meet the threshold with 76 and 84 respectively.

Discussion and conclusion

Financial distress is the period of accumulated difficulties, erupted tensions and conflict and a costly event that has the possibility of determining the firms' optimal capital structures. Financial distress is considered costly because of its potential tendency for organizations to act

in a way that are harmful to nonfinancial stakeholders and debtholders and weakened condition that induces an aggressive response by competitors that enhance an opportunity for gaining market share.

This study analyzes and presents the global trend of intellectuals' development of scholarly article on Global Research Trends in Predicting Financial Distress among Small and Medium-Sized Enterprises (SMEs) for the period of ten years 2014-2024 using Google Scholar data base. The study reveals that financial distress is an area with limited number of studies across the globe and Nigeria in particular. It further reveals the authors' strong collaboration and the keyword co-occurrence in the study of financial distress prediction under four clusters focusing on Model, Adaptation, Challenges, and Opportunities respectively.

Cluster three and four focusing on items such as Economic empowerment, poverty alleviation, and Islamic microfinance in Nigeria present limited studies which serve as the research direction for future studies. Similarly, cluster four also present another direction for further studies by presenting limited number of studies focusing on Economic empowerment, poverty alleviation, Islamic microfinance in Nigeria. In the same vein the keywords identified base on occurrences in the literature has the potential of guiding the future researcher as to the direction of the study area. This study further suggests for more studies on Bibliometric Analysis of Global Research Trends in Predicting Financial Distress among Small and Medium-Sized Enterprises (SMEs) using combinations of different databases to uncover more on the literature development on the study area of financial distress prediction among small and medium size enterprises.

References

- Altman, E. I. 1968, Financial Ratios, Discriminant Analysis, and the prediction of corporate Bankruptcy, *Journal of Finance*, Vol.23, No.4, 589-609.
- Åström, F. (2002). Visualizing library and information science concept spaces through keyword and citation based maps and clusters. In *The fourth international conference on Conceptions of Library and Information Science (CoLIS4)* (pp. 185-197). Libraries Unlimited.
- Ayvaz, R. N. (2023). A review on financial failure models- The case of manufacturing industry *Finansal başarısızlık modelleri üzerine bir inceleme - İmalat sanayi örneği*. 11, 375-399.
- Bahnson, P. R., & Bartley, J. W. (1992). The sensitivity of failure prediction models to alternative definitions of failure. *Advances in accounting*, 10, 255-278.
- Benckendorff, P., & Zehrer, A. (2013). A network analysis of tourism research. *Annals of tourism research*, 43, 121-149.
- Bichteler, J., & Eaton III, E. A. (1980). The combined use of bibliographic coupling and cocitation for document retrieval. *Journal of the American Society for Information Science*, 31(4), 278-282.
- Boadi, I., Sogah, E., Boadi, E. K., Gborse, F. C., Boadi, I., Sogah, E., Boadi, E. K., Christian, F., Boadi, I., Sogah, E., Boadi, E. K., Gborse, F. C., & Kwaku, J. (2024). Does monetary poverty reduction through gender empowerment work? *Cogent Social Sciences*, 10(1). <https://doi.org/10.1080/23311886.2024.2307696>
- Callon, M., Courtial, J. P., & Laville, F. (1991). Co-word analysis as a tool for describing the network of interactions between basic and technological research: The case of polymer chemistry. *Scientometrics*, 22(1), 155-205.
- Case, E. İ. (2022). *Sürdürülebilir Kırsal Kalkınmada Mikro Kredi'nin Kadın Girişimciliğine Etkisi : İzmir Örneği* / Impact of Microcredit in Sustainable Rural Development to Women 's. 11(1), 121-142. <https://doi.org/10.47130/bitlisos.1094559>
- Chang, Y. L. (2015). Examining relationships among elementary mathematics teacher efficacy and their students' mathematics self-efficacy and achievement. *Eurasia Journal of Mathematics, Science and Technology Education*, 11(6), 1307-1320.

- Cheng, X., Shuai, C., & Li, W. (2018). *The impact of eco-environment and disaster factors on poverty* :May. <https://doi.org/10.18402/resci.2018.04.03>
- Cheraghali, H., & Molnár, P. (2023). SME default prediction: A systematic methodology-focused review. *Journal of Small Business Management*, 00(00), 1–59. <https://doi.org/10.1080/00472778.2023.2277426>
- Cobb, G. (2021). *Ecosystem - based Adaptation in a Changing Climate : Systematic Review of its Limits* , Science and Knowledge. November 2020. <https://doi.org/10.13140/RG.2.2.12513.63843>
- Cobb, G. (2021). *Ecosystem - based Adaptation in a Changing Climate : Systematic Review of its Limits* , Science and Knowledge. November 2020. <https://doi.org/10.13140/RG.2.2.12513.63843>
- Cobo, M. J., López-Herrera, A. G., Herrera-Viedma, E., & Herrera, F. (2011). Science mapping software tools: Review, analysis, and cooperative study among tools. *Journal of the American Society for information Science and Technology*, 62(7), 1382–1402.
- Conference, I. S., & Politics, I. (2021). *Selected papers*.
- Culnan, M. J. (1986). The intellectual development of management information systems, 1972–1982: A co-citation analysis. *Management science*, 32(2), 156–172.
- De Bellis, Nicola (2009): *Bibliometrics and Citation Analysis*. From the Science Citation Index to Cybermetrics. Lanham: Scarecrow Press.
- Delaney, C. (1991). *The seed and the soil: Gender and cosmology in Turkish village society* (Vol. 11). Univ of California Press.
- Delgado, J. L. (2019). *Economic interventions and non-economic and economic variable associations with types of partner violence against women : A literature review Economic interventions and non-economic and economic variable associations with types of partner violence against*. February, 0–51.
- Dietrich, K. (1984). Joan Robinsons Konjunkturtheorie¹. *Wohlfahrt und Gerechtigkeit*, 2, 215.
- Dietrich, S. T., & Wagner, H. (1984). Critical surface scattering of X-rays at grazing angles. *Zeitschrift für Physik B Condensed Matter*, 56(3), 207–215.
- Ding, Z. L., Yu, Z. W., Yang, S. L., Sun, J. M., Xiong, S. F., & Liu, T. S. (2001). Coeval changes in grain size and sedimentation rate of eolian loess, the Chinese Loess Plateau. *Geophysical Research Letters*, 28(10), 2097–2100.
- Farias, F. Z., Valls, C., & Antonio, P. (2021). *Explanatory Factors of Business Failure : Literature Review and Global Trends*. 1–25.
- Financial, D., Through, I., & Networks, M. (2014). *Mobile Banking and Poverty Alleviation Master of Public Policy*.
- Format, J., Artikel, A., Zugang, F., German, J., Journal, I., Science, M., Holst, E., Meissner, A. T. M., Amsel, D. E. K., French, B. E. B., Parsons, G. B. J., Koev, G. B. D., Stanislav, B. G., Kornilova, C. Z. V., Aksenov, U. A. D., Bragin, U. A. V., Miros, R. U., Villa, P. L. D., Molteni, I. T. M., ... Lehmann, A. T. E. (n.d.). *ISSN (Print) 2701-8369 ISSN (Online) 2701-8377 Deutsche internationale Zeitschrift für zeitgenössische Wissenschaft of Modern Science*.
- Format, J., Artikel, A., Zugang, F., German, J., Journal, I., Science, M., Holst, E., Meissner, A. T. M., Amsel, D. E. K., French, B. E. B., Parsons, G. B. J., Koev, G. B. D., Stanislav, B. G., Kornilova, C. Z. V., Aksenov, U. A. D., Bragin, U. A. V., Miros, R. U., Villa, P. L. D., Molteni, I. T. M., ... Lehmann, A. T. E. (n.d.). *ISSN (Print) 2701-8369 ISSN (Online) 2701-8377 Deutsche internationale Zeitschrift für zeitgenössische Wissenschaft of Modern Science*.
- Foulkes, S. H. (2018). *Selected Papers*. *Selected Papers*. <https://doi.org/10.4324/9780429479847>
- Gangwani, D., & Zhu, X. (2024). Modeling and prediction of business success: a survey. *Artificial Intelligence Review*, 57(2), 44. <https://doi.org/10.1007/s10462-023-10664-4>
- Grosu, V., Chelba, A., Melega, A., Botez, D., & Socoliuc, M. (2023). Bibliometric analysis of the literature on evaluation models of the bankruptcy risk. *Strategic Management*, 28(2), 21–44. <https://doi.org/10.5937/straman2200035g>
- Gupta, A., Dengre, V., Kheruwala, H. A., & Shah, M. (2020). Comprehensive review of text-mining applications in finance. *Financial Innovation*, 6(1), 1–25. <https://doi.org/10.1186/s40854-020-00205-1>
- Hanif, M., Hassan, A., & Yahya, S. (2017). *determinants of repayment behavior in islamic microfinance, the role of accountability as mediator and customer service as moderator : a case study of bank kerjasama rakyat malaysia*.
- Hauser, M., & Mugonya, J. (2024). Framework for conceptualising transition readiness from emergency response to rebuilding livelihoods in Mogadishu , Somalia. *Food Security*, 397–409. <https://doi.org/10.1007/s12571-024-01431-6>

- Jacinta, C., Paper, A. W., & Wp, N. (2018). *Recent finance advances in information technology for inclusive development : A systematic review University of Central Lancashire*.
- Kasali, T. A. (2020). *Influence of Microfinance Intervention on Rural Poverty Alleviation in South-west Nigeria : An Application of Propensity Score Matching Technique*. 411–422.
- Kassab, E. A., Nordin, N., Amlus, M. H., & Ahmad, B. (2022). Drivers for SMEs Sustainability: A Review and Research Agenda. *Journal of Positive School Psychology*, 2022(12), 1520–1540. <http://journalppw.com>
- Kelimeler, A. (2022). *İslami Mikrofinans : Bir Literatürİncelemesi Islamic Microfinance : A Literature Review*. 12(1), 169–210. <https://doi.org/10.31679/adamakademi>.
- Kim, H., Cho, H., & Ryu, D. (2020). Corporate default predictions using machine learning: Literature review. *Sustainability (Switzerland)*, 12(16), 6325. <https://doi.org/10.3390/SU12166325>
- Korsakien, R. (2022). *Sustainability Drivers of Small and Medium-Sized Firms : A Review and Research Agenda*. 30(1), 1–12. <https://doi.org/10.46585/sp3001380>
- Kura, S. S., Viswanathan, K. K., & Ishak, S. (2017). *Microsavings : A Substitute Or A Complement To Microcredit In The Fight Against*. 7, 174–179.
- Leydesdorff, L. (1997). Why words and co-words cannot map the development of the sciences. *Journal of the American society for information science*, 48(5), 418–427.
- Mabasa, N. C. (2017). *rsityveTo w n versity e To w*.
- Marqués, A. I., García, V., & Sánchez, J. S. (2020). Ranking-based MCDM models in financial management applications: analysis and emerging challenges. *Progress in Artificial Intelligence*, 9(3), 171–193. <https://doi.org/10.1007/s13748-020-00207-1>
- Matenda, F. R., Sibanda, M., Chikodza, E., & Gumbo, V. (2021). Bankruptcy prediction for private firms in developing economies: a scoping review and guidance for future research. *Management Review Quarterly*, 1–40.
- Melega, A. (2022). Bibliometric analysis of scientific production regarding the harmonization of accounting in BRICS emerging economies. *European Journal of Accounting, Finance & Business*, 10(1), 11–20.
- Ohlson, J. (1980). "Financial Ratios and the Probabilistic Prediction of Bankruptcy," *Journal of Accounting Research*, 19, 109–131
- Peralungal, A., & Natchimuthu, N. (2024). *Machine Learning in Financial Distress : A Scoping Review Alay Peralungal *, NatchimuthuNatchimuthu*. 5(March), 457–474. <https://doi.org/10.47857/irjms.2024.v05i03.0779>
- Peters, M. (2016). *BUNDLING OF BUSINESS DEVELOPMENT SERVICES AS A MEANS TO OVERCOME*. April.
- Pritchard, A. 1969. "Statistical Bibliography or Bibliometrics?" *Journal of Documentation* 25, no. 4:348–49
- Ramdani, R., Ratnasari, R. T., & Mawardi, I. (2022). *Management and Strategy in Waqf Institution : A Systematic Literature Review Manajemen Dan Strategi Lembaga Wakaf : Tinjauan Pustaka yang Sistematis*. 6(2), 135–145. <https://doi.org/10.21070/perisai.v6i2>.
- Rixtel, A. Van, Ioana, A., Kimie, H., Alexopoulou, I., & Harada, K. (2003). The new Basel capital accord and its impact on Japanese banking: A qualitative analysis. *New Basel Capital Accord*. <http://cei.ier.hit-u.ac.jp/activities/seminars/papers/Adrian.pdf>
- Sathyanarayana, N. (2024). *A systematic review of models for the prediction of corporate insolvency Revisiónsistemática de los modelos de predicción de la insolvenciaempresarial*. <https://doi.org/10.56294/sctconf2024952>
- Sathyanarayana, N., & Narayanan, R. (2024). A systematic review of models for the prediction of corporate insolvency. *Salud, Ciencia y Tecnología - Serie de Conferencias*, 3. <https://doi.org/10.56294/sctconf2024952>
- Scott, W. R. (1981). Developments in organization theory, 1960–1980. *American Behavioral Scientist*, 24(3), 407–422.
- Sirelkhatim, M. A., & Salim, N. (2015). Prediction of Banks Financial Distress. *SUST Journal of Engineering and Computer Sciences (JECS)*, 16(1), 40–55.
- Small, H. (1973). Co-citation in the scientific literature: A new measure of the relationship between two documents. *Journal of the American Society for information Science*, 24(4), 265–269.
- Small, H. G. (1978). Cited documents as concept symbols. *Social studies of science*, 8(3), 327–340.

- Soukal, I., Mačí, J., Trnková, G., Svobodova, L., Hedvičáková, M., Hamplova, E., Maresova, P., & Lefley, F. (2024). A state-of-the-art appraisal of bankruptcy prediction models focussing on the field's core authors: 2010–2022. *Central European Management Journal*, 32(1), 3–30. <https://doi.org/10.1108/CEMJ-08-2022-0095>
- Su, H.-N., & Lee, P.-C. (2010). Mapping knowledge structure by keyword co-occurrence: a first look at journal papers in technology foresight. *Scientometrics*, 85(1), 65–79.
- Van Eck, N. J., & Waltman, L. (2014). Visualizing bibliometric networks. In *Measuring scholarly impact: Methods and practice* (pp. 285–320). Cham: Springer International Publishing.
- Van Raan, A. F. (2005). For your citations only? Hot topics in bibliometric analysis. *Measurement: interdisciplinary research and perspectives*, 3(1), 50–62.
- View metadata, citation and similar papers at core.ac.uk.* (2016).
- Vogel, R., & Güttel, W. H. (2013). The dynamic capability view in strategic management: A bibliometric review. *International Journal of Management Reviews*, 15(4), 426–446.
- Whittaker, J. (1989). Creativity and conformity in science: Titles, keywords and co-word analysis. *Social studies of science*, 19(3), 473–496.
- Wu, Shinong, and Xianyi Lu. "A study of models for predicting financial distress in China's listed companies. " *Economic Research Journal* 6(2001):46–55.
- Zhao, Y., Du, H., Li, Q., Zhuang, F., Liu, J., & Kou, G. (2022). A Comprehensive Survey on Enterprise Financial Risk Analysis from Big Data Perspective. *ArXiv Preprint ArXiv:2211.14997*.
- Zmijewski, M. E. (1984). Methodological issues related to the estimation of financial distress prediction models. *Journal of Accounting research*, 59–82.